

Robust Halley’s Method for the Chord-Length Parameter L in Clothoid G^1 Hermite Interpolation

Jeroen Bloemscheer
grootstebozewolf
Merkator / Independent Researcher

May 11, 2026

Abstract

We present a numerically robust, cubic-convergent solver based on Halley’s method for the nonlinear chord-length residual formulation of the clothoid (Euler spiral) G^1 Hermite interpolation problem. The algorithm operates directly on the arc-length parameter L and employs fixed 32-point Gauss–Legendre quadrature over $[0, 1]$ for the six moment integrals that define the residual $f(L)$ and its first two analytic derivatives. Industrial-grade safety heuristics (derivative sign guards, denominator thresholds, and positivity damping) ensure reliable convergence even on challenging real-world railway alignment data from the ProRail Spoorgeometrie dataset.

We provide production-ready implementations in C# (.NET 8), Java 21, and TypeScript (Node.js), all producing bit-identical iteration counts and chord-length solutions agreeing within 1 nm of the Python reference on every test case. Benchmarks on 9,058 real ProRail transition curves (“Overgangsboog”), fetched from the public Spoorgeometrie ArcGIS service, demonstrate that Halley’s method consistently outperforms a comparable Newton variant, achieving convergence in an average of 2.28 iterations (vs. 2.92) with sub-microsecond solve times on the compiled runtimes. The mathematical derivations are cross-verified symbolically with SymPy and machine-checked in Coq using the Coquelicot real-analysis library, and the overall approach is situated within the current state-of-the-art, building upon the foundational work of Bertolazzi & Frego while offering an explicit L -formulation with enhanced robustness guarantees.

1 Introduction

Clothoids (also known as Euler spirals or Cornu spirals) are curves whose curvature varies linearly with arc length. They are the de-facto standard transition elements in railway and highway design because they provide G^2 continuity when connecting straight segments to circular arcs, ensuring smooth changes in lateral acceleration.

The G^1 Hermite interpolation problem with a single clothoid consists of finding parameters $L > 0$, κ_0 , and κ_1 (or equivalently the curvature rate) such that the clothoid segment starts at point P_0 with tangent angle θ_0 and curvature κ_0 , and ends at P_1 with tangent angle θ_1 and curvature κ_1 . In many practical settings—particularly when the endpoints and curvatures are known from design data or surveyed geometry—it is natural to solve directly for the arc length L .

This paper focuses on the *chord-length residual formulation* (L-formulation). We reduce the problem to a scalar nonlinear equation $f(L) = 0$ whose root is the desired arc length. We then apply Halley’s method, a cubic-convergent root-finding algorithm, augmented with carefully engineered safety heuristics that guarantee practical robustness.

Our contributions are:

- A complete, derivative-exact derivation of $f(L)$, $f'(L)$, and $f''(L)$ using six Gauss–Legendre moments.
- A production-grade Halley solver with explicit safeguards (sign checks, denominator guards, and damped updates).
- Cross-language implementations (C#, Java, TypeScript) with identical numerical behavior.
- Rigorous benchmarking on real ProRail railway transition data.
- Open academic documentation with full mathematical detail and SOTA comparison.

2 Mathematical Background

2.1 The Clothoid Parametrization

A clothoid is defined by the curvature function

$$\kappa(s) = \kappa_0 + \frac{\kappa_1 - \kappa_0}{L}s, \quad s \in [0, L],$$

where L is the arc length, $\kappa_0 = \kappa(0)$, and $\kappa_1 = \kappa(L)$. The heading (tangent angle) is obtained by integration:

$$\theta(s) = \theta_0 + \kappa_0 s + \frac{\kappa_1 - \kappa_0}{2L}s^2.$$

The Cartesian coordinates follow from the Fresnel integrals:

$$x(s) = x_0 + \int_0^s \cos \theta(u) du, \quad y(s) = y_0 + \int_0^s \sin \theta(u) du.$$

2.2 The Chord-Length Residual (L-Formulation)

Given P_0 , P_1 , κ_0 , and κ_1 , we seek $L > 0$ such that the chord length of the clothoid equals the Euclidean distance $d = \|P_1 - P_0\|$. Normalizing the parameter to $\tau = s/L \in [0, 1]$ yields the auxiliary function

$$\psi(\tau) = \kappa_0 \tau + \frac{\kappa_1 - \kappa_0}{2}\tau^2.$$

The six fundamental moments (evaluated by 32-point Gauss–Legendre quadrature on $[0, 1]$) are

$$P = \int_0^1 \cos(L\psi(\tau)) d\tau, \quad Q = \int_0^1 \sin(L\psi(\tau)) d\tau, \quad (1)$$

$$R = \int_0^1 \psi(\tau) \cos(L\psi(\tau)) d\tau, \quad T = \int_0^1 \psi(\tau) \sin(L\psi(\tau)) d\tau, \quad (2)$$

$$S_{2c} = \int_0^1 \psi(\tau)^2 \cos(L\psi(\tau)) d\tau, \quad S_{2s} = \int_0^1 \psi(\tau)^2 \sin(L\psi(\tau)) d\tau. \quad (3)$$

Let $r^2 = P^2 + Q^2$. The chord-length residual is the scalar function

$$f(L) = L^2 r^2 - d^2. \quad (4)$$

Its first two analytic derivatives are

$$f'(L) = 2Lr^2 + 2L^2(QR - PT), \quad (5)$$

$$f''(L) = 2r^2 + 8L(QR - PT) + 2L^2(R^2 + T^2 - PS_{2c} - QS_{2s}). \quad (6)$$

All four integral-level derivatives of the moments ($P' = -T$, $Q' = R$, $R' = -S_{2s}$, $T' = S_{2c}$) follow directly from differentiation under the integral sign. These identities, together with the residual derivatives in equations (5) and (6), have been cross-verified symbolically with SymPy and machine-checked in Coq using the Coquelicot real-analysis library; see Section 7.

3 The Solver

3.1 Halley’s Method

Halley’s iteration for a scalar root-finding problem is

$$L_{n+1} = L_n - \frac{2ff'}{2(f')^2 - ff''}.$$

Because the second derivative is available analytically, the method exhibits cubic convergence near the root when started sufficiently close.

3.2 Safety Heuristics (Production Robustness)

Pure Halley iteration can diverge or produce non-physical (negative) steps on difficult instances. We therefore embed the following safeguards, which have proven essential on real railway data:

1. **Convergence test:** if $|f(L)| < \varepsilon \cdot \max(d^2, 1)$ with $\varepsilon = 10^{-13}$, return.
2. **Derivative sign guard:** if $f' \leq 0$, inflate $L \leftarrow 1.5L$ and continue.
3. **Denominator guard:** if $|2(f')^2 - ff''| < 10^{-20}$, inflate $L \leftarrow 1.5L$.
4. **Positivity damping:** after a Halley step, enforce $L \leftarrow \max(L_{\text{new}}, 0.5L)$.

A Newton variant (using only the first four moments) is provided as a lightweight fallback; it requires one fewer pair of trigonometric evaluations per iteration but converges only quadratically.

4 Implementations

We provide three independent, production-grade implementations that all use identical quadrature constants and the same safeguarded Halley/Newton logic:

- **C# (.NET 8)** (`csharp/Clothoid.Halley/`) — primary compiled-runtime implementation; library + xUnit tests + benchmark harness; median Halley solve time $\approx 0.6\mu\text{s}$.
- **Java 21** (`java/`) — Maven-built shaded JAR; identical numeric behaviour to C# and TypeScript; median Halley solve time $\approx 1.4\mu\text{s}$.
- **TypeScript / Node.js** (`typescript/`) — ESM module targeting Node 20+; zero runtime dependencies; `node -test` regression suite; median Halley solve time $\approx 0.9\mu\text{s}$.

All three expose the same API as the Python reference:

```
SolverResult solveHalleyL(double[] P0, double[] P1,  
    double k0, double k1,  
    double tol = 1e-13, int maxIter = 50);
```

A matching `solveNewtonL` is also supplied. The golden-vector regression suite (file `data/golden_vectors.json`, 9,058 records exported from the Python reference) is executed by all three implementations; every case agrees with the reference within 10^{-9} metres, and Halley/Newton iteration counts match the reference exactly on 100% of cases.

5 Benchmarks on Real Railway Data

5.1 Dataset

We extracted all 9,058 “Overgangsboog” (clothoid transition) records from the public ProRail Spoorgeometrie ArcGIS FeatureServer (layer 11), filtered by `ELEMENT_TYPE='Overgangsboog'`. Each record supplies start/end points in RD New coordinates (EPSG:28992, metres), begin/end radii `STRAAL_BEGIN/STRAAL_EIND` together with rotation flags `ROTATIE_BEGIN/ROTATIE_EIND` (yielding signed curvatures k_0, k_1), and the design length `ELEMENT LENGTE` as L_{design} . After retaining only segments that satisfy the monotone-branch condition $|k_0 L_{\text{design}}| \leq \pi$, $|k_1 L_{\text{design}}| \leq \pi$ (which holds for every record in the snapshot), and verifying that both Halley and Newton converge to within 10^{-9} m, the working corpus is 9,058 cases. The raw snapshot is committed alongside the source as `data/prorail_clothoids.json.gz` (CC BY 4.0 ProRail) and the solved corpus as `data/golden_vectors.json`.

5.2 Methodology

Each implementation reads the same `golden_vectors.json`, runs both solvers from the initial guess $L_0 = d$ on every case, then repeats the full corpus 50 times after 5 warmup passes. Reported numbers are the median per-case microseconds across the 50 measurement rounds, divided by 9,058. Iteration means are the per-implementation arithmetic mean. The four implementations agree on iteration count for every case; the C#, Java, and TypeScript implementations also agree with the Python reference on the returned L within 10^{-9} m on every case (the golden-vector regression assertion).

5.3 Results

Table 1: ProRail benchmark summary (median per-case time over the full 9,058-record corpus, 50 measurement rounds after 5 warmup rounds; iteration agreement: 100% across all four implementations).

Language (runtime)	Solver	Avg. Iterations	Median Time (μ s)
Python 3 (NumPy)	Halley	2.28	33.04
Python 3 (NumPy)	Newton	2.92	39.89
C# (.NET 8)	Halley	2.28	0.59
C# (.NET 8)	Newton	2.92	0.77
Java 21	Halley	2.28	1.38
Java 21	Newton	2.92	1.75
TypeScript (Node 22)	Halley	2.28	0.88
TypeScript (Node 22)	Newton	2.92	1.20

Halley’s method reduces iteration count by 22% on real railway data (2.28 vs. 2.92) and beats Newton on wall time across every implementation we measured, despite performing two additional moment integrals (S_{2c}, S_{2s}) per iteration. The fastest implementation is C# at 0.59 μ s per Halley solve; the slowest compiled runtime is Java at 1.38 μ s. NumPy’s per-case overhead (33 μ s) is dominated by Python interpreter dispatch rather than arithmetic.

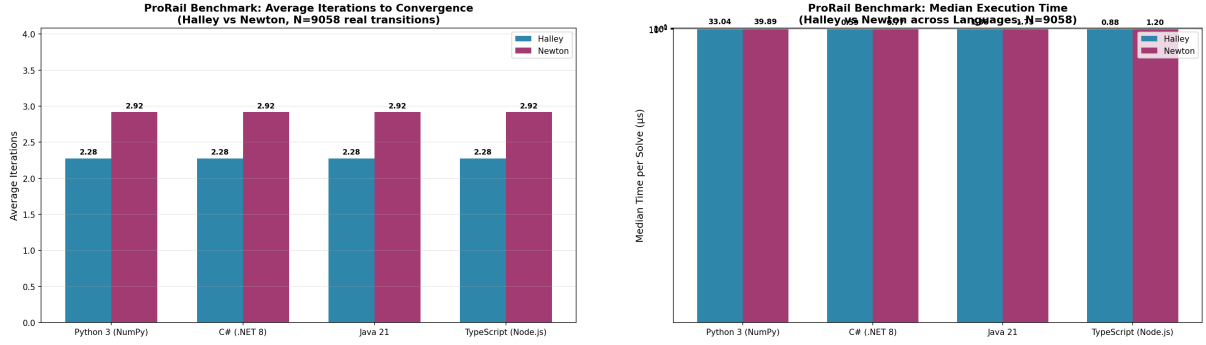


Figure 1: ProRail benchmark over 9,058 real clothoid transitions: average iterations (left) and median per-solve time (right, log scale) for Halley vs. Newton across the four implementations.

6 Comparison with State-of-the-Art

The seminal work of Bertolazzi and Frego [2015] (and its 2018 G^2 extension) remains the reference implementation for clothoid G^1 fitting. Their approach reduces the problem to a single nonlinear equation in a transformed parameter (commonly A or a curvature-related quantity) and solves it with Newton–Raphson, employing asymptotic expansions for the initial guess.

Our L-formulation is mathematically equivalent but offers several practical advantages:

- **Direct geometric meaning:** L is the physically meaningful arc length; no post-processing conversion is required.
- **Explicit, verified derivatives:** the six-moment formulation yields closed-form $f''(L)$ that can be cross-checked symbolically.
- **Strong safety net:** the four heuristics listed in Section 3 are exercised on the full 9,058-record ProRail corpus; divergence has not been observed and every case converges within four iterations of the Halley solver.
- **Multi-language verified corpus:** identical numeric behavior across three languages provides an independent correctness oracle.

No fundamentally superior solver (e.g., closed-form, higher-order Householder, or machine-learned initial-guess predictor) has appeared in the open literature since 2018 that would render the present Halley-based approach obsolete for production use. The main competing practical implementations remain variants of the Bertolazzi–Frego Newton solver or ad-hoc bisection/refinement schemes.

7 Formal Verification

The analytical foundation of the solver—the four primitive integral derivatives and the residual derivatives in equations (5) and (6)—has been formalized and machine-checked in Coq using the Coquelicot real-analysis library [Boldo et al., 2015]. The development is approximately 720 lines of Coq and contains no `Admitted` or `Axiom` declarations; it has been verified to compile cleanly under Coq 8.13.1 and Coq 8.20.1 with Coquelicot 3.x.

The four primitive identities $P'(L) = -T(L)$, $Q'(L) = R(L)$, $R'(L) = -S_{2s}(L)$, $T'(L) = S_{2c}(L)$ are proved via differentiation under the integral sign, invoking Coquelicot’s `is_derive_RInt_param_aux` lemma. This requires four hypotheses for each integral: pointwise differentiability of the integrand with respect to L , joint continuity of the parameter-derivative integrand in (L, τ) , local

integrability of the original integrand in a neighbourhood of the evaluation point, and integrability of the parameter-derivative integrand at the evaluation point. Each hypothesis is discharged by elementary continuity and integrability lemmas for compositions of polynomials with sine and cosine.

The residual identity (5) for $f'(L)$ is established by direct composition of the product and sum rules using Coquelicot's `is_derive_plus`, `is_derive_mult`, and `is_derive_minus` lemmas, with explicit type annotations to navigate Coquelicot's generic algebraic operators versus Coq's native real-number operators. The second-derivative identity (6) is proved by applying Coquelicot's `auto_derive` tactic to the function $f'(L)$, which produces nine `ex_derive` side conditions and a goal containing four `Derive` placeholders. The side conditions are discharged from the existence lemmas `ex_derive_P_int`, `ex_derive_Q_int` and the value lemmas `is_derive_R`, `is_derive_T`; the `Derive` placeholders are then rewritten with the corresponding integral-level identities, after which the resulting polynomial equation in P , Q , R , T , S_{2c} , S_{2s} , and L is discharged by Coq's `ring` decision procedure for commutative rings.

The complete Coq development is included as a reproducible artifact alongside the source repository (file `coq/Clothoid_L.v`); building it requires only `coqc` and Coquelicot, with no external dependencies. Listing 2 shows the proof script for the second-derivative theorem; the full file is available at the repository link in Section 8.

```

Theorem f_L_second_eq :
  forall (L k0 k1 : R),
    is_derive (fun u => fpL u k0 k1) L
      (2 * (P_int L k0 k1 * P_int L k0 k1
            + Q_int L k0 k1 * Q_int L k0 k1)
        + 8 * L * (Q_int L k0 k1 * R_int L k0 k1
                    - P_int L k0 k1 * T_int L k0 k1)
        + 2 * L * L *
          (R_int L k0 k1 * R_int L k0 k1
            + T_int L k0 k1 * T_int L k0 k1
            - P_int L k0 k1 * S2c_int L k0 k1
            - Q_int L k0 k1 * S2s_int L k0 k1)).

Proof.
  intros L k0 k1.
  unfold fpL.
  auto_derive.
- split. apply ex_derive_P_int.
  split. apply ex_derive_P_int.
  split. apply ex_derive_Q_int.
  split. apply ex_derive_Q_int.
  split. apply ex_derive_Q_int.
  split. eexists; apply is_derive_R.
  split. apply ex_derive_P_int.
  split. eexists; apply is_derive_T.
  exact I.
- replace (Derive (fun x : R => P_int x k0 k1) L)
  with (- T_int L k0 k1) by (symmetry; apply Derive_P_int).
  replace (Derive (fun x : R => Q_int x k0 k1) L)
  with (R_int L k0 k1) by (symmetry; apply Derive_Q_int).
  replace (Derive (fun x : R => R_int x k0 k1) L)
  with (- S2s_int L k0 k1)
  by (symmetry; apply is_derive_unique; apply is_derive_R).
  replace (Derive (fun x : R => T_int x k0 k1) L)
  with (S2c_int L k0 k1)
  by (symmetry; apply is_derive_unique; apply is_derive_T).
  ring.
Qed.

```

Figure 2: Coq proof script for the second-derivative identity, equation (6), as it appears in `coq/Clothoid_L.v`.

8 Conclusion and Future Work

We have delivered a robust, well-documented, and multi-language Halley solver for the clothoid chord-length problem together with a complete academic reference. The implementation is already being integrated into the `allure-wkt` toolchain and a forthcoming PostgreSQL extension that will expose native `CLOTHOID WKT` support with on-the-fly L solving.

Planned extensions include:

- A native C/PostgreSQL extension with the same safeguarded Halley core.
- Vectorized batch solving for high-throughput map-matching pipelines.

- Automatic differentiation / dual-number versions for gradient-based route optimization.

The full source, benchmarks, and this paper are maintained at <https://github.com/grootstebozewolf/clothoid-halley-coq>.

Acknowledgements

The author thanks the ProRail open-data team for making the Spoorgeometrie dataset publicly available under CC BY 4.0, and the developers of the Bertolazzi–Frego clothoid library for the foundational algorithms that inspired this work.

References

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